

Muscle Tissues

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Department: physiology &
Microbiology**

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Objectives:

- know the structure, location and characteristic feature of each type of muscle.

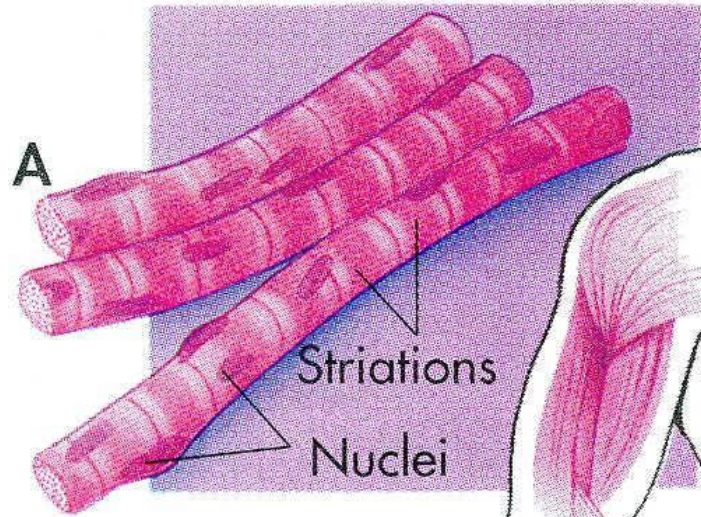
Muscle tissue

- Muscle tissue is one of the four basic tissues.
- Muscle cells are contractile cells.
- Specialized proteins (actin & myosin) allow the cells to contract.
- Their ability to contract provides a mechanism for movement of the internal organs

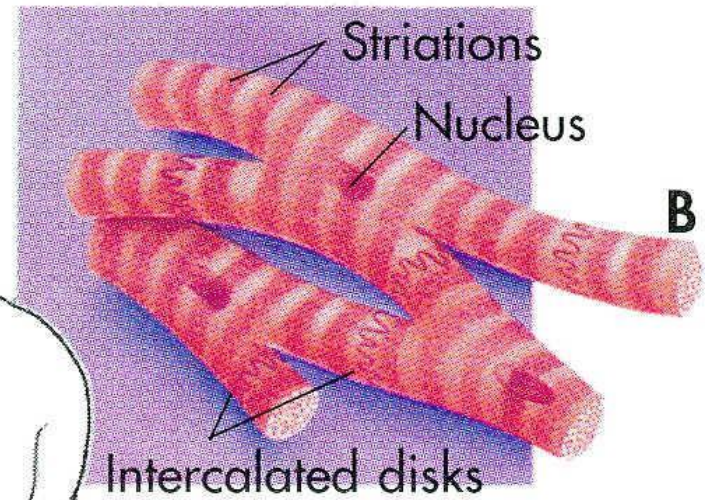
Muscle tissue can be divided into 3 major types:

- Skeletal, Cardiac and Smooth muscle.
- Skeletal muscle: is attached to bones.
- Cardiac muscle: is found in the heart.
- Smooth muscle: is found in vessels, ducts, skin, and internal organs.
- Muscle tissue is may be controlled by nerves, hormones, local chemicals, or itself depending on the type and location.

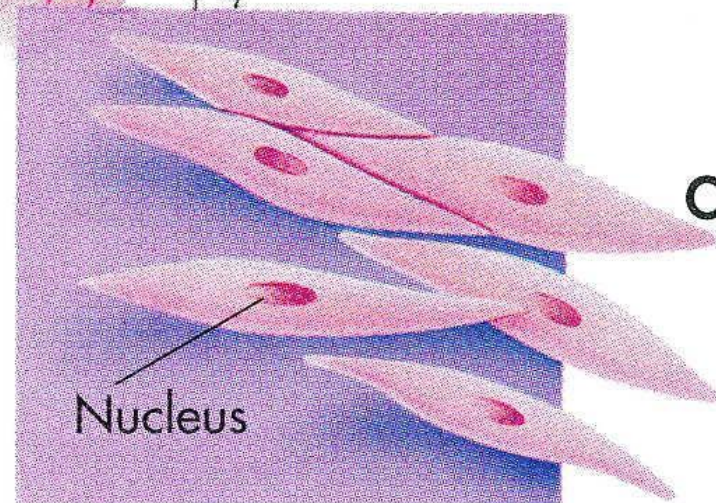
Skeletal muscle fibers



Cardiac muscle fibers

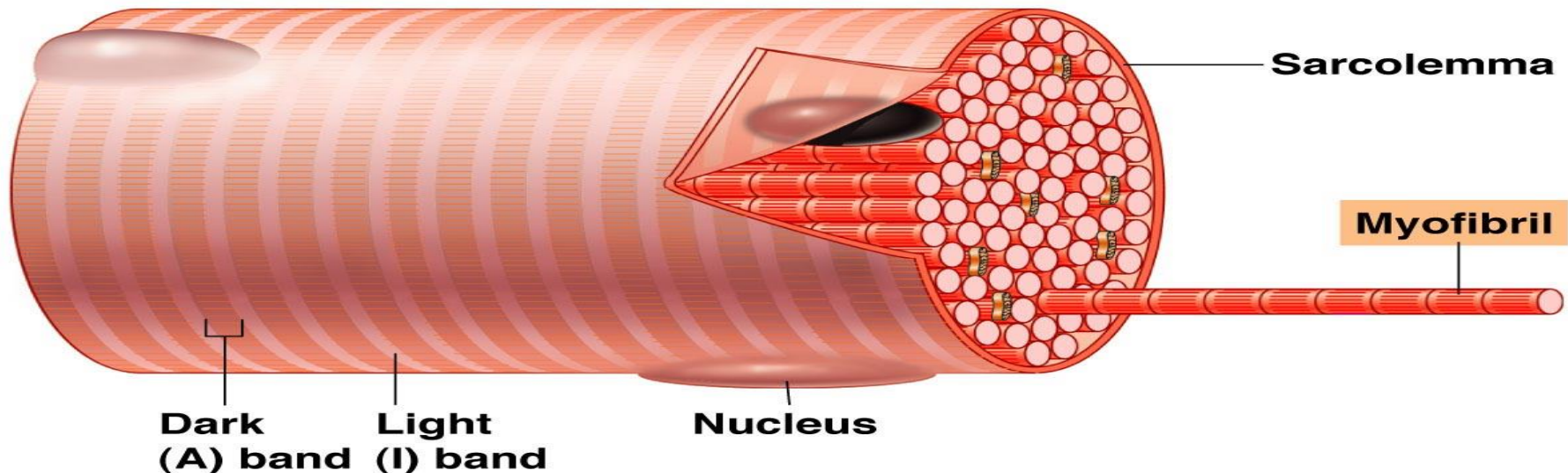


Smooth muscle fibers



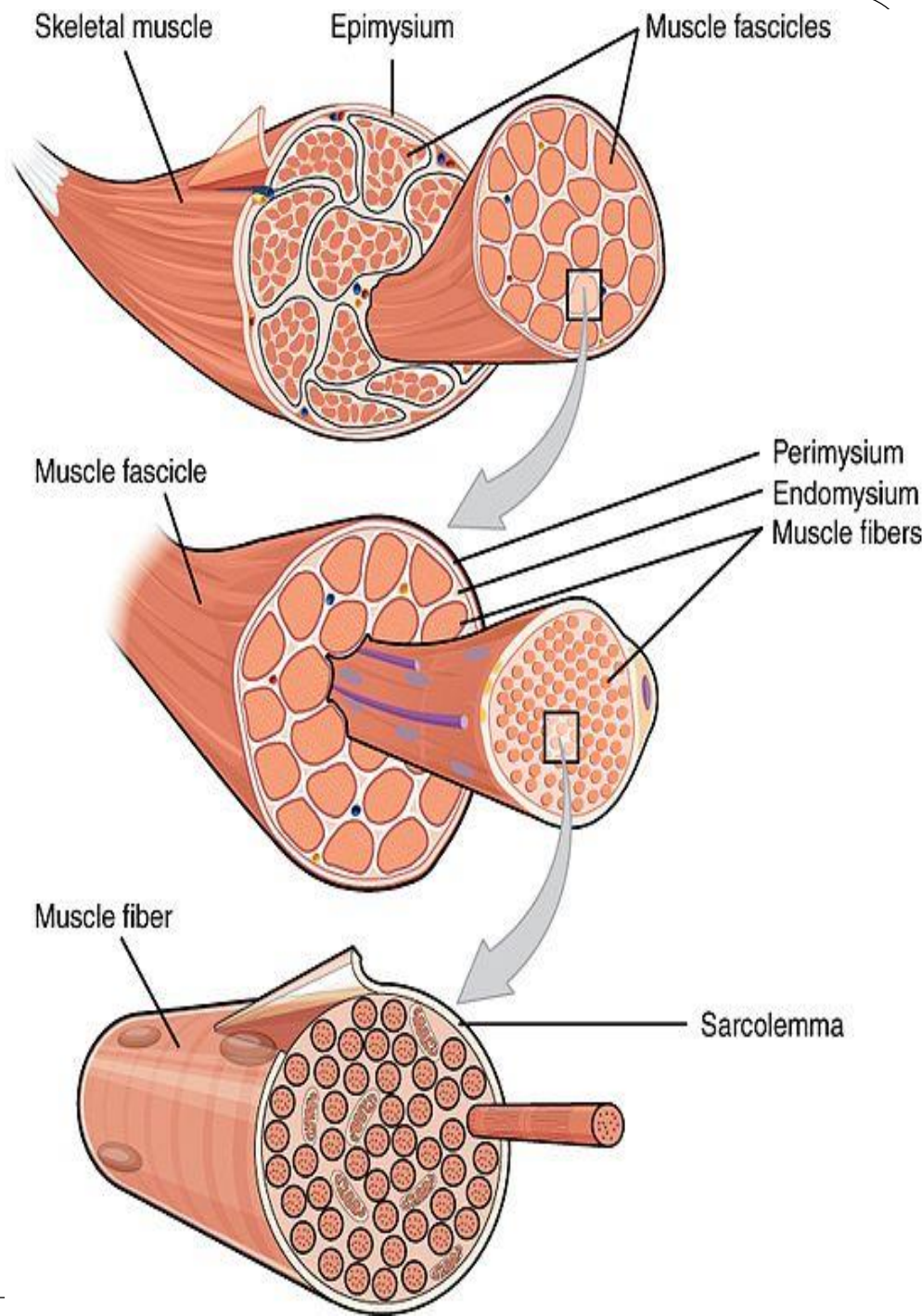
Skeletal muscle

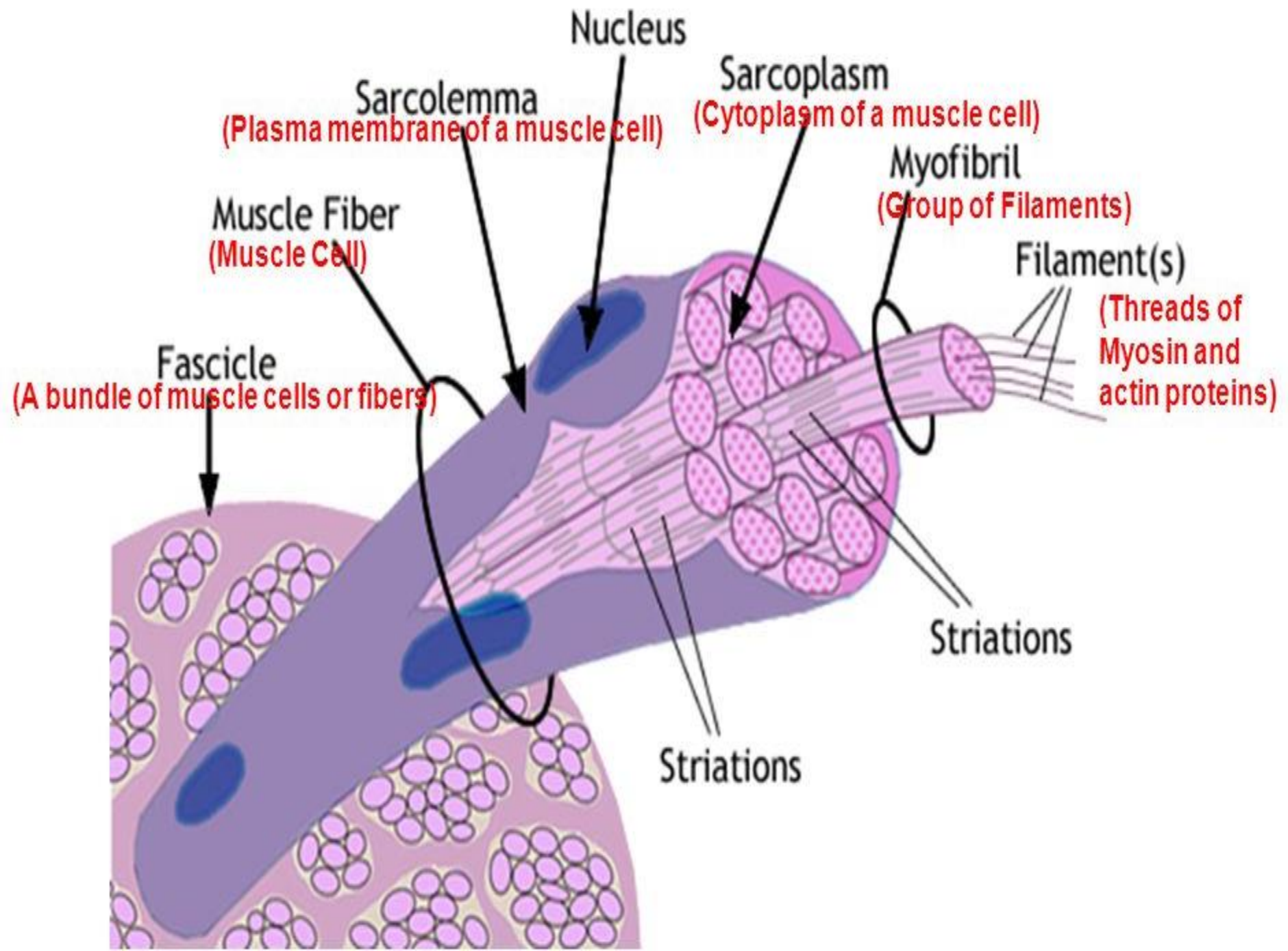
- under voluntary control.
- attaches to bones.
- the muscle cells (muscle fibers) are long, unbranched, multinucleated and arranged in parallel bundles.
- Muscle fibers are composed of bundles of myofibrils.



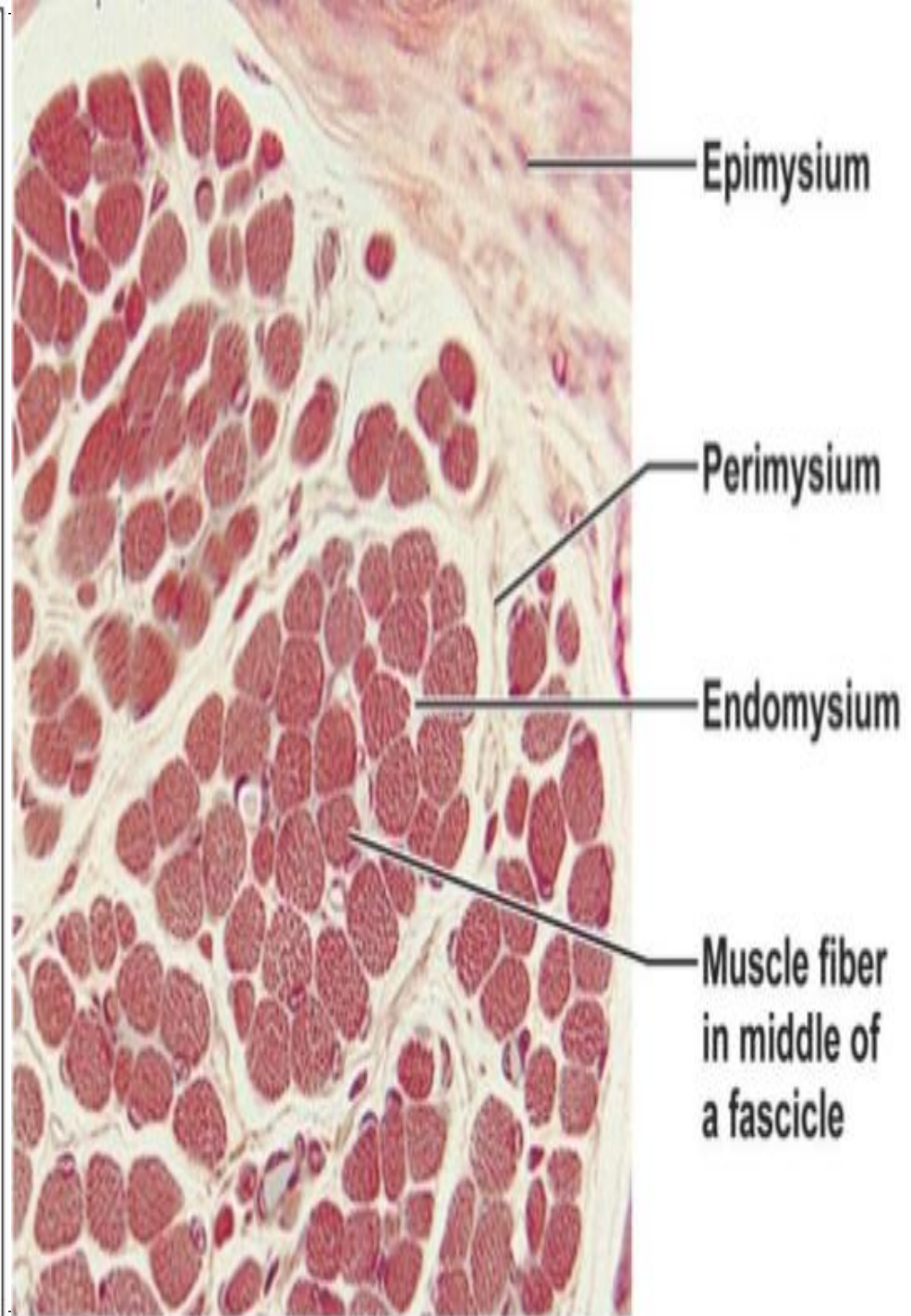
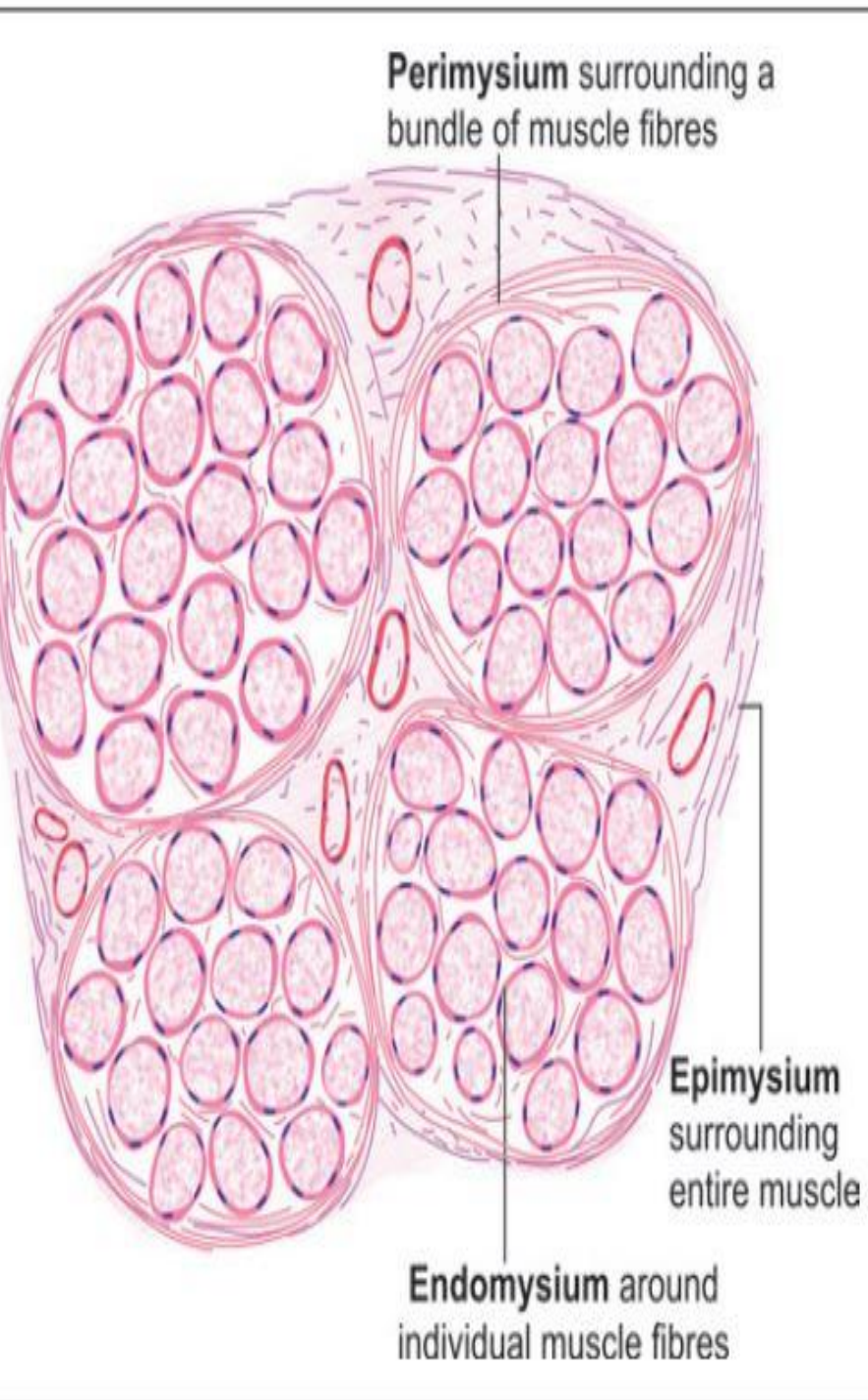
(a) Segment of a **muscle fiber** (cell)

- Each skeletal muscle composed of regular muscle bundles.
- A number of muscle bundles or fascicles grouped to form an entire muscle .
- Number of muscle fibers grouped to form muscle fascicle.

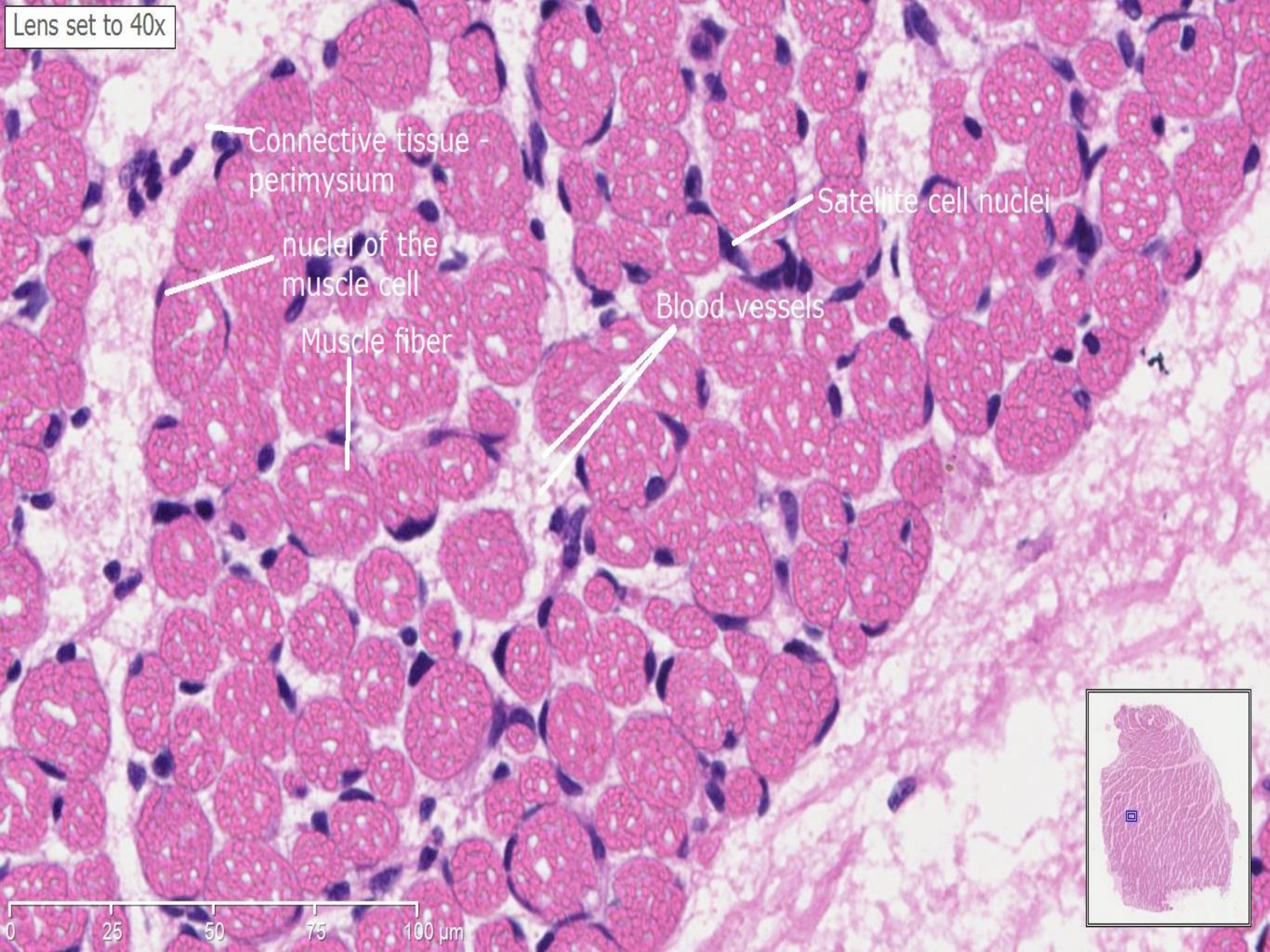




- The entire skeletal muscle is surrounded by a sheath of dense connective tissue called **epimysium**,
- from which there are thin septa extending to separate between the muscle bundles or fascicles called **perimysium**.
- Finally each individual muscle fiber which is called myofibrils is surrounded by a delicate connective tissue sheath called the **endomysium**.
- These septa carry blood vessels that nourish the muscles & carry their contraction.



Lens set to 40x



Connective tissue -
perimysium

nuclei of the
muscle cell

Muscle fiber

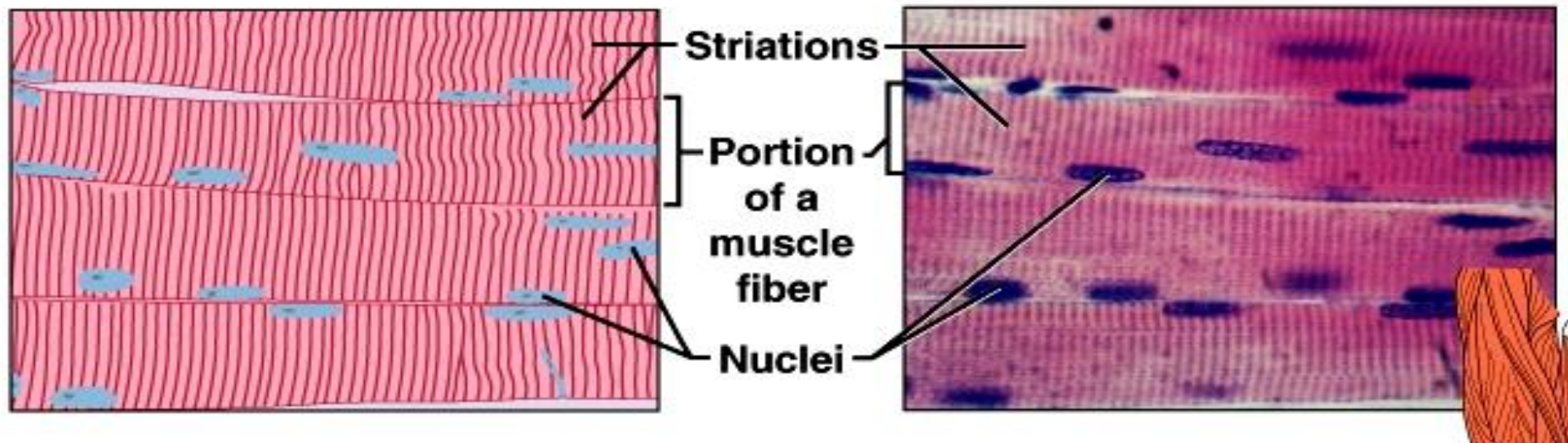
Satellite cell nuclei

Blood vessels

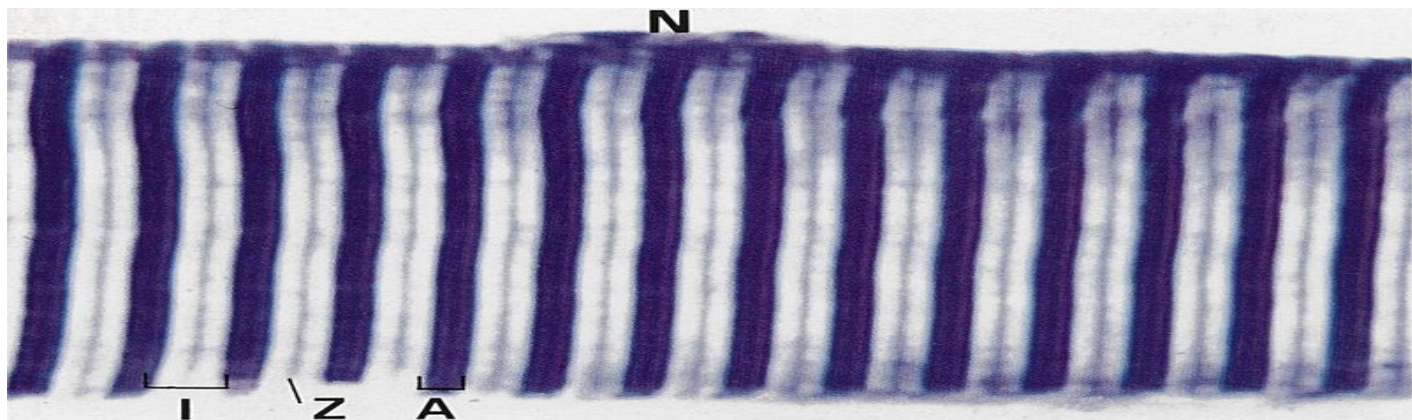
0 25 50 75 100 μm



- By light microscopy, the cells of both cardiac and skeletal muscle show the presence of transverse striations. For this reason, both of these muscle types are referred to as "striated."
- The striated appearance results from the presence of a highly organized network of thin filaments (actin) and Thick filaments (myosin).

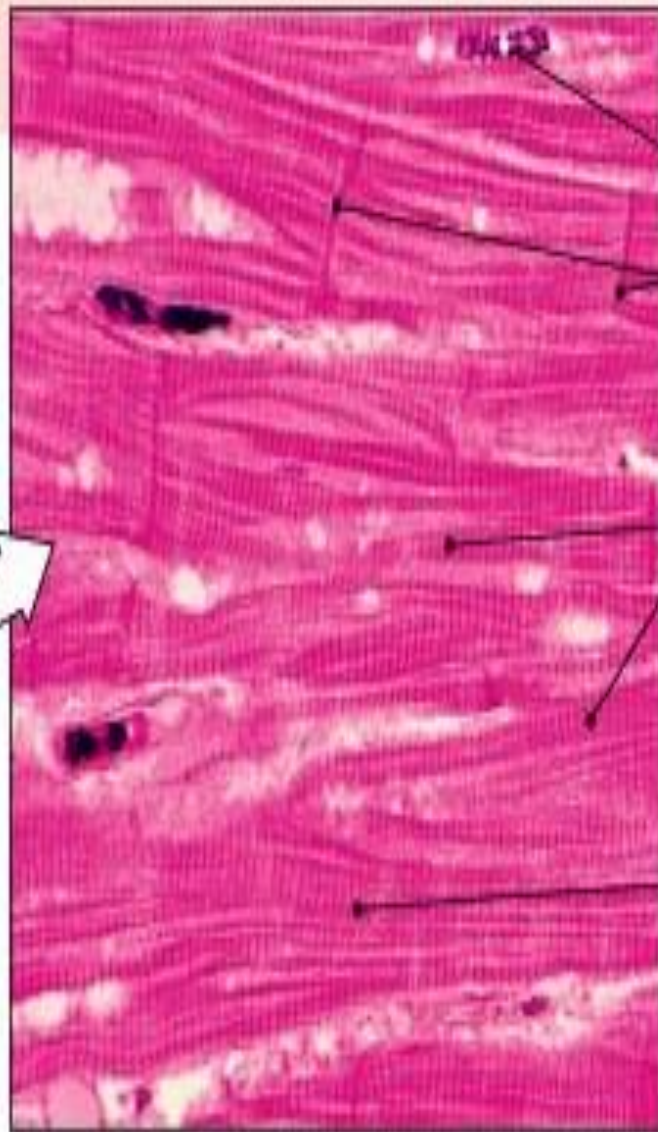


- Myofibrils are cylindrical structures within muscle fiber, consists bundles of protein filaments called **myofilaments** (actin & myosin)
- Myofibril when shortens, muscle shortens (contracts)
- The arrangement of these myofilaments is responsible for the muscle striation, which is in longitudinal section of muscle fiber under light M. present as alternating (light & dark) bands.



Cardiac muscle

- Is involuntary, striated in appearance and contracts rhythmically and automatically.
- It is found in the myocardium (the muscle layer of the heart).
- Adjacent cardiac cells joined via gap junctions, these junctions appear as dark lines called **intercalated disc**.
- Cardiac muscle cells (fibers) are short, branched and interconnected, has a single, centrally located nucleus.
- These junctions allow the spread of excitation to pass from one fiber to another



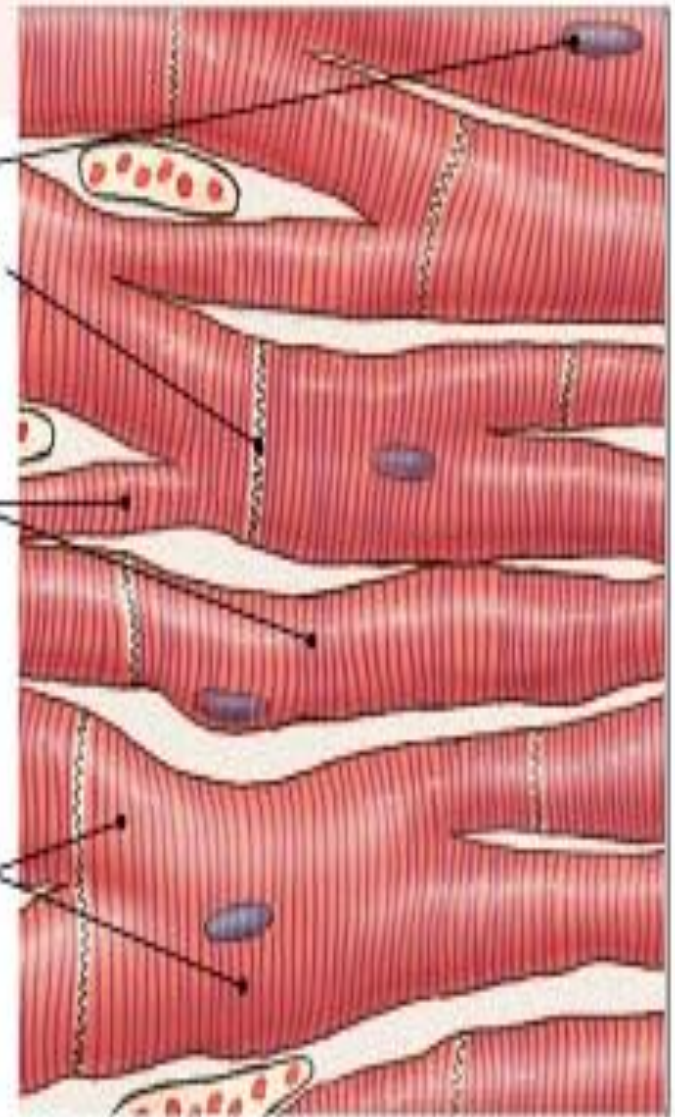
LM $\times 450$

Nucleus

Intercalated
discs

Cardiac
muscle
cells

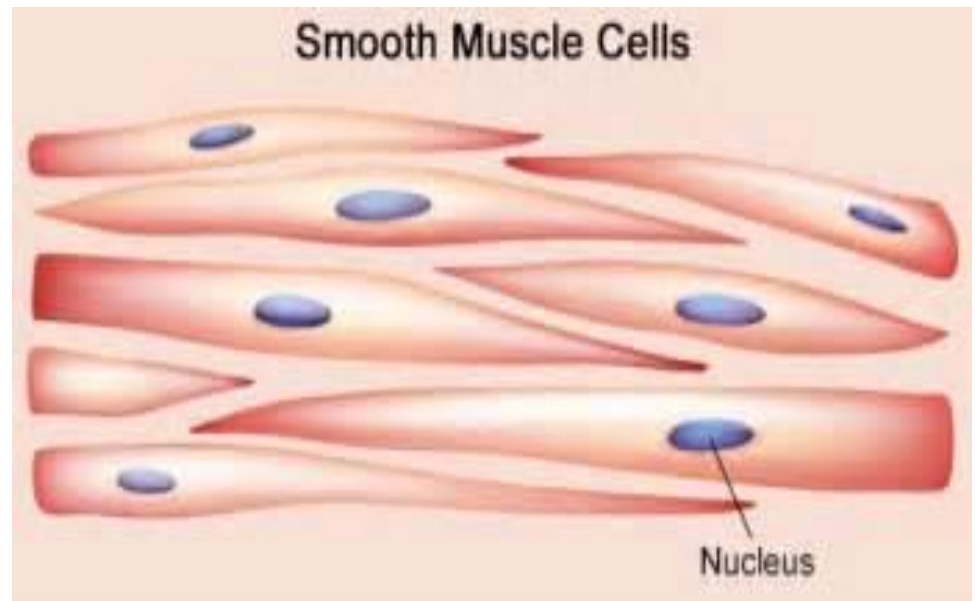
Striations



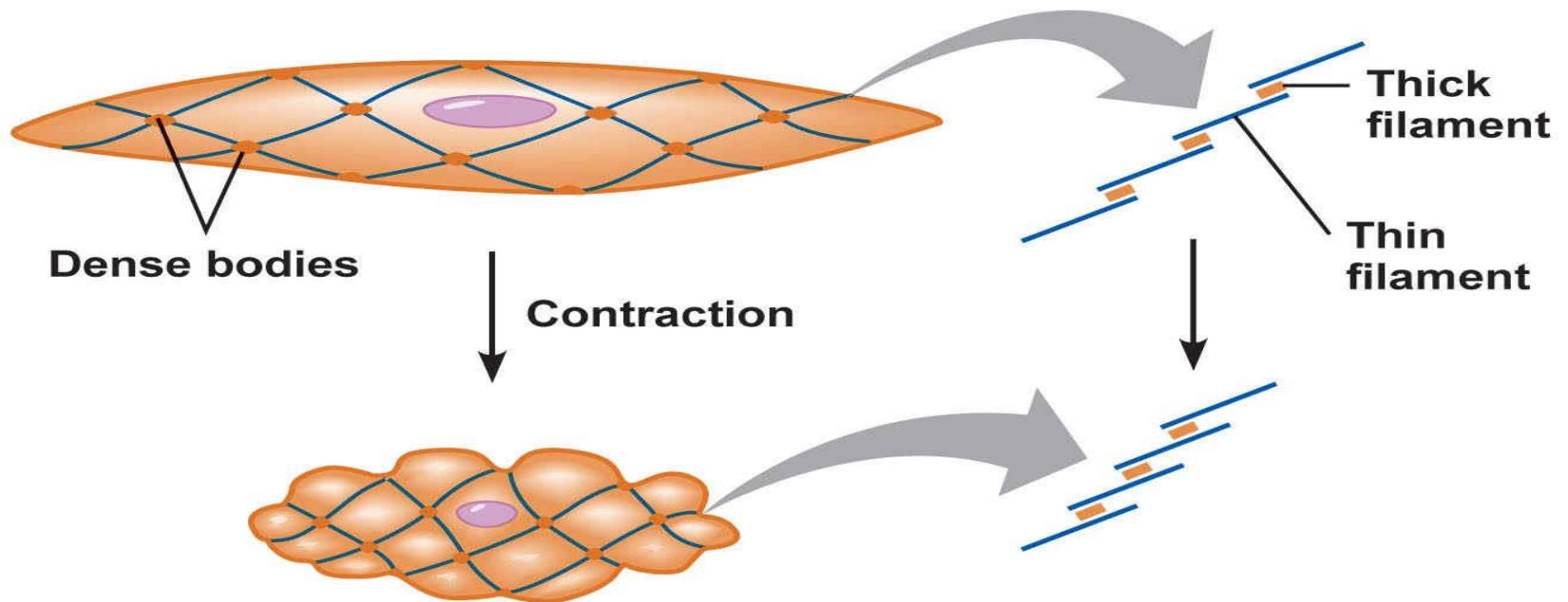
(b) Cardiac muscle

Smooth muscle

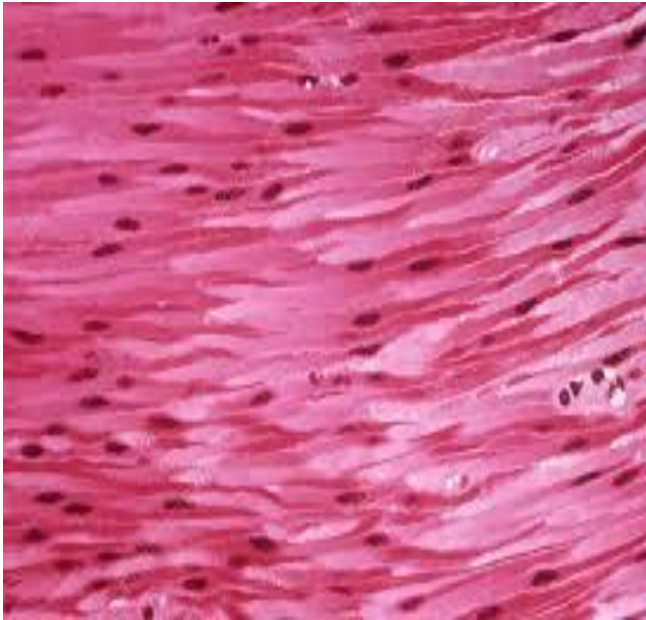
- involuntary control and found in the walls of blood vessels, digestive, respiratory, urinary, and reproductive organ
- Are small and spindle shaped cells with a central nucleus. They do not branch, and there are no striations.



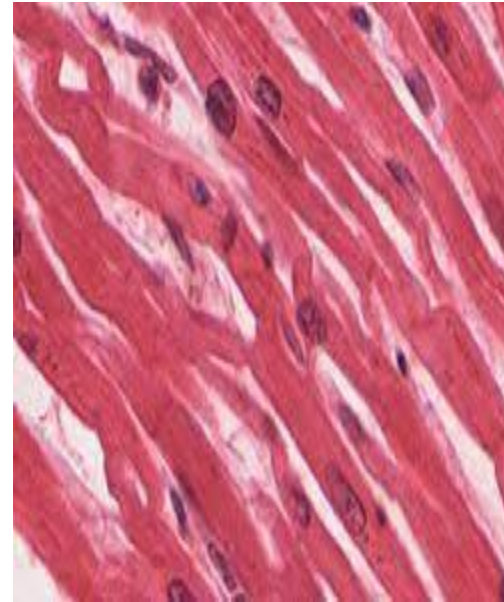
- Smooth muscle fibers contain numerous myofibrils that are oriented along the long axis of the fiber, and which extend from end to end within the fiber. These myofibrils are composed of the same myofilaments of **actin** and **myosin** contained in the other muscle tissues, except that they are arranged in a more random fashion.



Smooth muscle



Cardiac muscle



Skeletal muscle

